



DESN2000 SENG

Part 2 Group Report

LaTeX Framework

Project title	<i>[Insert project title]</i>
Team members	<i>[Insert team member names and zIDs]</i>
Submission date	<i>[Insert submission date]</i>

Contents

1	Introduction and Part 2 Scope	1
2	Problem Statement and Research Justification	1
2.1	Problem Statement	1
2.2	Research Justification	1
3	Solution Ideation	2
3.1	Research-to-Ideation Transition	2
3.2	Generated Technical Concepts	3
3.3	Ideation Outcome	5
3.4	Ideation Method	5
4	Concept Selection and Prioritisation	6
4.1	Selection Method	6
4.2	Decision Matrix	7
4.3	Selected Concept and Rationale	8
5	Product Backlog Structure	9
5.1	Epic Overview	9
5.2	User Stories	10
6	Acceptance Criteria	12
6.1	EP-01 Housemate Discovery and Trust	12
6.2	EP-02 Shared Household Living	14
6.3	EP-03 Accountability Extensions	19
6.4	EP-04 Property Management	20
7	Functional and Non-Functional Requirements	21
7.1	Functional Requirements	21
7.2	Non-Functional Requirements	21
8	User Story Dependencies	22
8.1	Dependency Types	22
8.2	Dependency Table	22
8.3	Dependency Graph	24
9	First-Pass User Story Map	25
9.1	Sprint 1 Scope and Rationale	25

10 Preliminary Prototype	26
10.1 Prototype Scope	26
10.2 Prototype Flow	26
10.3 Prototype Screens	26
11 Prototype-to-User-Story Traceability	27
12 Prototype Test Plan	27
13 Conclusion	30
14 References	31
A Full User Story and Use Case Mapping	31
B Prototype Screens and Links	33
C Jira Evidence	33

1 Introduction and Part 2 Scope

Part 1 of this project used surveys and interviews across four stakeholder groups to build personas and to define the problem that a shared-living platform should solve. This report continues from that foundation and applies the Ideate stage of the design process from an end-user perspective, as the Preliminary Report brief requires. Every design decision made here is justified against the Part 1 evidence rather than introduced as a new assumption.

The report is organised along the path from evidence to testable prototype. Section 2 restates the approved problem and the research that justifies the design direction. Sections 3 and 4 show how the findings were turned into three candidate concepts and how the strongest concept was selected systematically. Sections 5 to 9 define the product backlog, in which Epics are broken down into user stories with measurable acceptance criteria, supported by a dependency analysis and a first-pass story map that defines Sprint 1. Sections 10 to 12 present the preliminary HTML prototype, trace each screen back to the stories it demonstrates and set out the plan for testing it. The backlog itself is submitted in Jira, and every story table in this report links to its ticket.

2 Problem Statement and Research Justification

2.1 Problem Statement

The problem statement approved in Part 1 is restated below in its agreed form.

Create a mobile platform which allows university students and shared accommodation residents to seamlessly manage household responsibilities, expenses and housemate communication, while also enabling property managers to streamline tenancy administration and tenant communication, to reduce conflict between tenants, streamline property management, and improve the overall shared living experience.

The statement names the target users, describes the problems they experience and locates where in the shared-living journey support is needed. It spans two connected stages. Before move-in, users must find trustworthy accommodation and compatible housemates. After move-in, they must manage shared responsibilities without the conflict that informal arrangements tend to produce. The concept selected in Section 4 addresses exactly this span, and every user story in the backlog is written against one of these two stages.

2.2 Research Justification

The design direction follows directly from the Part 1 research. Table 1 links the strongest evidence to the persona it affects, the implication it creates for the problem and the design response adopted in this report. Only the evidence needed to justify the direction is repeated here, and Section 3.1 shows how the same findings were carried into ideation.

Table 1. Research-to-solution traceability.

Research evidence	Persona or user need	Problem implication	Solution implication
Every one of the 16 international respondents reported concern about rental scams, and 43.75% had encountered misleading or false listing information.	International student persona, unfamiliar with the Australian rental market.	Users cannot judge whether a listing or a stranger is genuine before first contact.	Verification with visible trust indicators on listings and profiles (US-11, US-13).
87.5% rated housemate compatibility very or extremely important, and 68.75% had experienced conflict with previous housemates.	Emily Chen, the long-term housemate persona, who needs housemates with similar daily expectations.	Incompatible daily habits surface only after move-in, when they are hardest to resolve.	Structured lifestyle profile with compatibility comparison (US-12).
44% named communication difficulties as the hardest part of the search, and over 80% were willing to use preference-based matching.	International student persona, who fears exposing personal details to strangers.	Early contact through personal channels exposes users to scams and harassment.	Guided in-app messaging that hides personal contact details (US-09).
All surveyed domestic students relied on verbal household agreements, all had experienced conflict over responsibilities, and 66.67% wanted better chore tracking.	Domestic student persona, managing shared living for the first time.	Verbal expectations break down after move-in and cause avoidable conflict.	Recorded house rules and agreements supported by shared responsibility tools (US-02, US-10, EP-02).

3 Solution Ideation

3.1 Research-to-Ideation Transition

The ideation process began by translating the key findings of the Interim Report and Design Journal 1 into technical opportunities. Rather than generating ideas in isolation, the team first restated the strongest evidence from the user research so that every subsequent concept could be traced back to a validated user need. Table 2 summarises the findings that carried the most weight in this transition.

Table 2. Key research findings carried forward into ideation.

Research finding	Evidence from user research
Rental scam concern	All 16 international student respondents (100%) reported concern about rental scams.
Misleading listing information	43.75% of respondents had experienced misleading or false accommodation information.
Housemate compatibility	87.5% of respondents considered housemate compatibility very or extremely important. The most frequently selected compatibility factors were quiet and respectful rest time, smoking and drinking habits, cleanliness, daily schedule, and willingness to share household work.
Informal shared-living expectations	The domestic shared-living research, while based on a smaller sample, showed that verbal household agreements, unclear bill arrangements and weak accountability could lead to conflict after move-in.

These findings were then grouped into four recurring research themes, and each theme was reframed as a design need that a software solution could address. This mapping, shown in Table 3, formed the shared starting point for the ideation activities described in Section 3.4: it ensured that the ideas generated by the team responded to evidenced user problems rather than assumed ones, and it later served as a reference when screening and prioritising the candidate concepts.

Table 3. Mapping of research themes to design needs.

Research theme	Design need
Trust and scam concern	Verification and visible trust indicators
Housemate compatibility	Structured lifestyle profile and compatibility comparison
Unsafe or unclear communication	In-app messaging and guided discussion prompts
Informal shared-living expectations	Recorded co-living agreement and reminders

Framing the four design needs as a single design challenge produced the guiding ideation question for this stage of the project:

How might we help students and young renters move from an uncertain housemate search to a safer, informed and clearly agreed co-living arrangement?

All ideation activities in the remainder of this section were run against this question, so that every generated concept could be assessed on how well it advanced users along that journey.

3.2 Generated Technical Concepts

Working from the four design needs and the guiding ideation question, the team generated three candidate concepts. Rather than producing minor variations of one idea, each concept was deliberately anchored on a different part of the user journey, so that the later selection stage could compare genuinely distinct design directions. Table 4 summarises the three concepts, and Tables 5–7 describe each concept in detail.

Table 4. Overview of the three generated concepts.

Concept	Name	Journey focus	Design needs addressed
A	Verified Contact Gateway	Trust and safe first contact	Verification and trust indicators; safe in-app communication
B	Explainable Compatibility Matcher	Comparison of practical daily-living habits	Structured lifestyle profile and compatibility comparison
C	Trust-to-Agreement Co-living Journey	End-to-end flow from search to an agreed co-living arrangement	All four design needs

Table 5. Concept A — Verified Contact Gateway.

Aspect	Description
Focus	Trust and safe first contact between unfamiliar users.
Core features	Student, email or identity verification; visible verification badges; in-app messaging that does not expose phone numbers or social-media accounts; report and block functions; an inspection and listing authenticity checklist.
Technical mechanism	A verification state machine stores each verification type as <i>unverified</i> , <i>pending</i> , <i>verified</i> , <i>rejected</i> or <i>expired</i> . The profile interface displays only verification outcomes, never the raw identity documents.
Strength	Directly addresses the universal scam concern found in the research and users’ reluctance to contact unknown people.
Limitation	Confirms that a user has passed a check, but does not determine whether two people are suitable long-term housemates.

Table 6. Concept B — Explainable Compatibility Matcher.

Aspect	Description
Focus	Helping users compare practical daily-living habits before committing.
Core features	Structured lifestyle questionnaire; non-negotiable compatibility filters; weighted compatibility score; match-reason explanation; prompts for discussing important differences.
Technical mechanism	The system compares attributes such as cleanliness, quiet hours, smoking, daily schedule, guests, privacy and chore expectations. Candidates who violate mandatory constraints are removed first, and a weighted similarity score is then calculated for the remainder.
Strength	Responds directly to the research finding that compatibility is practical and grounded in daily routines rather than only personality or demographics; Design Journal 1 likewise concluded that users need compatibility questions and comparison indicators.
Limitation	A high compatibility score does not guarantee that users will clearly discuss bills, chores and household boundaries before moving in.

Table 7. Concept C — Trust-to-Agreement Co-living Journey.

Aspect	Description
Focus	Combining verification, compatibility assessment, guided communication and agreement creation into one end-to-end flow.
Core features	Structured housemate profile creation; verification of selected account information; lifestyle compatibility comparison; guided in-app chat; creation of a shared co-living agreement; reminders for accepted responsibilities.
Technical mechanism	Separate subsystems manage profile data, verification status, compatibility calculation, guided prompts, agreement versioning, reminders and privacy permissions, communicating through defined service events such as <i>ProfileUpdated</i> , <i>MutualMatchCreated</i> and <i>AgreementAccepted</i> .
Strength	Supports the complete journey represented by the Emily Chen persona — searching online, inspecting and verifying, then moving into shared living — which Design Journal 1 identified as requiring verification, safe chat, compatibility and co-living management across all three stages.
Limitation	Has the largest technical scope of the three concepts and therefore requires a carefully controlled MVP.

Together, the three concepts cover an increasing share of the user journey: Concept A secures the first contact, Concept B improves the quality of the match, and Concept C carries the users through to a recorded agreement. This deliberate spread of scope and risk provided a meaningful basis for the systematic screening and prioritisation described in Section 4.

3.3 Ideation Outcome

Comparing the three concepts against the guiding ideation question showed how far each advanced users along the target journey. Concept A addresses trust, but not compatibility or future shared-living expectations. Concept B improves the housemate decision, but does not fully support users after a match is made. Concept C addresses the research-supported relationship between pre-move-in uncertainty and post-move-in conflict, connecting the earlier stages of the journey to the recorded co-living agreement that the other two concepts leave unaddressed. Table 8 summarises this comparison.

Table 8. Comparison of concept coverage against the user journey.

Concept	Coverage of the target journey	Outcome
A	Addresses trust, but not compatibility or future shared-living expectations.	Not carried forward on its own
B	Improves the housemate decision, but does not fully support users after a match.	Not carried forward on its own
C	Addresses the research-supported link between pre-move-in uncertainty and post-move-in conflict across the full journey.	Taken forward for evaluation

Concept C was therefore taken forward for the systematic evaluation described in Section 4.

3.4 Ideation Method

Required content

Name and explain the ideation method used by the team. Show how the method started from the problem definition and produced multiple ideas. Include evidence from the actual ideation activity.

Table 9. Ideation method template.

Ideation stage	Method used	Team activity	Output
<i>[Stage 1]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>
<i>[Stage 2]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>
<i>[Stage 3]</i>	<i>[Method]</i>	<i>[What the team did]</i>	<i>[Produced artefact]</i>

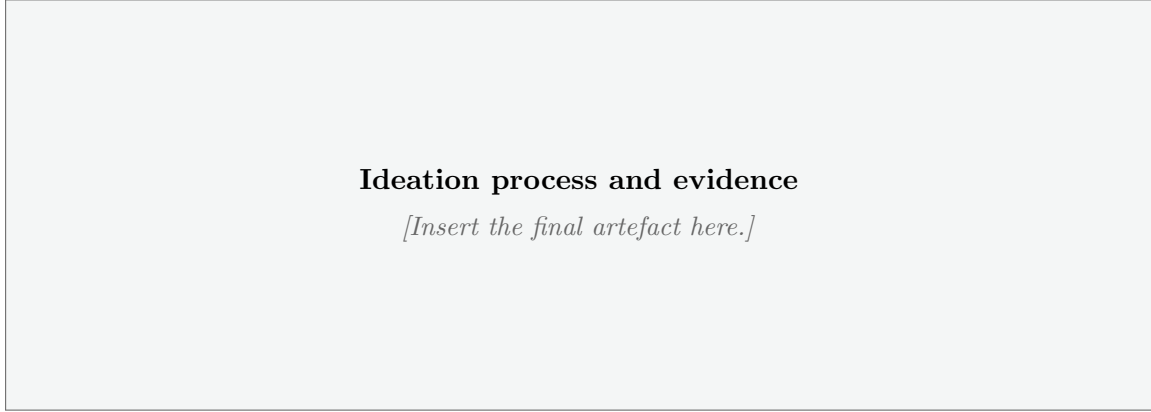


Figure 1. Ideation process and evidence.

4 Concept Selection and Prioritisation

4.1 Selection Method

The team used a two-stage prioritisation method so that selecting a solution concept and controlling its implementation scope were treated as separate decisions. First, a weighted decision matrix compared the three concepts generated in Section 3.2 against common research, user-value, delivery and validation criteria, rather than selecting a concept on preference alone. Second, after Concept C had been selected, MoSCoW prioritisation defined its minimum viable product (MVP), because selecting the most complete concept does not imply that every associated feature can be implemented in the first prototype. The matrix therefore answered *which concept should be pursued*, and MoSCoW answered *which parts of that concept must be delivered first*.

Each selection criterion was traced to the research findings, the assessment requirements or the project constraints, as shown in Table 10.

Table 10. Origin and rationale for the concept-selection criteria.

Criterion	Weight	Origin and weighting rationale
Research evidence fit	30%	The Preliminary Report requires the ideation process and resulting user stories to be justified by the problem definition, user research and personas. This criterion received the highest weight because a concept that cannot be traced to the research is not an appropriate solution, however convenient it is to prototype.
User value / risk reduction	20%	Measures how far each concept reduces the four research-supported risks in Table 3: uncertain trust, poor lifestyle fit, unsafe communication and unclear shared-living expectations — the direct value delivered to the primary users.
Feasibility in a 10-week prototype	20%	The project is constrained to a ten-week schedule and requires a working prototype with room for iteration. Weighting feasibility equally with user value prevents an ambitious concept from being selected without recognising its delivery risk.
Prototype testability	15%	The marking rubric requires the preliminary prototype to be usable for testing the features represented by the user stories. This criterion assesses whether the concept can be expressed as observable tasks and a coherent clickable prototype.
Innovation beyond existing tools	10%	The project brief asks for an innovative, user-centred solution. This criterion distinguishes concepts that reproduce an existing isolated tool from those that create a new, research-justified user journey.
Accessibility / privacy manageability	5%	Accessibility and privacy are explicit project considerations. The low numerical weight reflects that this criterion also acts as an acceptance gate: a concept with an unacceptable accessibility or privacy risk would be rejected regardless of its weighted total.

Each team member first rated the three concepts independently on a five-point scale (1 = fails the criterion; 3 = partial satisfaction with material limitations; 5 = fully satisfies the criterion under the agreed MVP assumptions). The independent ratings were then compared in the Week 6 team meeting: where ratings differed, each member stated the evidence, assumption or delivery concern behind the score before a consensus rating was recorded. The final matrix therefore documents a team judgement rather than an unexamined average (Figure 2).



Figure 2. Week 6 concept-selection evidence: independent ratings and consensus discussion.

4.2 Decision Matrix

Each concept's weighted total was calculated by multiplying its rating on every criterion by that criterion's weight and adding the results. Because the six weights add up to 100%, the maximum possible total is 5.00. As an example, Concept C's rating of 5 on research evidence fit contributes

$5 \times 30\% = 1.5$ points to its total of 4.35. Table 11 presents the result.

Table 11. Weighted decision matrix for the three generated concepts.

Selection criterion	Weight	Concept A	Concept B	Concept C
Research evidence fit	30%	4	4	5
User value / risk reduction	20%	4	4	5
Feasibility in a 10-week prototype	20%	5	4	3
Prototype testability	15%	4	5	4
Innovation beyond existing tools	10%	3	4	5
Accessibility / privacy manageability	5%	4	4	3
Weighted total	100%	4.10	4.15	4.35

Concept C achieved the highest weighted score (4.35/5), ahead of Concept B (4.15/5) and Concept A (4.10/5). Its advantage came from its stronger fit with the research evidence, broader end-to-end user value and greater innovation beyond a single-purpose tool.

The scoring was deliberately conservative rather than assigning Concept C the highest rating on every criterion. Concept A received the highest feasibility rating because its verification and safe-contact flow has the narrowest implementation boundary. Concept B received the highest testability rating because profile comparison and compatibility explanation can be tested as a self-contained interaction. Concept C received the lowest feasibility and privacy-manageability ratings because it introduces dependencies between profile data, verification, compatibility, guided communication and agreement creation, and contains more consent and information-disclosure decisions. The matrix therefore makes Concept C’s implementation risk visible instead of concealing it within a post-hoc justification.

4.3 Selected Concept and Rationale

Why Concept C was selected. Concept C, the *Trust-to-Agreement Co-living Journey*, was selected because it is the only concept that addresses the full research-supported chain from pre-move-in uncertainty to post-move-in conflict. Concept A reduces scam and unsafe-contact risk through verification and protected communication, but does not let users evaluate lifestyle compatibility or record household expectations. Concept B supports a more informed housemate decision, but stops before users negotiate and preserve shared rules. As shown in Table 8, Concepts A and B each address one part of the user journey, whereas Concept C links trust assessment, lifestyle comparison, safe discussion and explicit agreement in one coherent flow — which explains its higher research-fit and user-value scores in Table 11.

Managing the selected concept’s weakness. The principal limitation of Concept C is its lower feasibility within the available prototype schedule. MoSCoW prioritisation was therefore used to prevent the selected concept from becoming an uncontrolled commitment to implement every possible feature. Features were classified by whether they were necessary to test the core problem statement, whether other user stories depended on them, and whether they could be represented meaningfully in the MVP prototype. Table 12 translates the selected concept directly into prioritised user stories.

Table 12. MoSCoW prioritisation for the Concept C MVP.

Priority	Mapped user stories and features	Scope rationale
Must have	Profile and lifestyle comparison (US-12); verification indicators (US-11); safe in-app communication (US-09); basic co-living agreement creation and confirmation (US-02, US-10).	These stories form the minimum end-to-end path required to test whether a student can assess trust, compare living expectations, communicate without disclosing personal contact details, and record an agreed expectation before commitment.
Should have	Agreement-based reminders (US-14); expense logging and bill splitting (US-04–US-06).	These features strengthen post-agreement follow-through, but the core Trust-to-Agreement assumption can be evaluated without them in the first clickable prototype.
Could have	Housemate review and reputation system (US-15).	A review system adds a further trust signal but requires moderation rules, sufficient participation and careful treatment of bias; it is not necessary for validating the first-sprint journey.
Won't have in the MVP	Property-manager-side functionality (US-16–US-18).	These stories introduce a separate stakeholder role, additional permissions and new data dependencies. Excluding them keeps the MVP focused on the student-to-student journey; they remain candidates for a later release.

Only the Must-have stories define the first-sprint prototype path. Should-have stories remain available for a subsequent iteration, while Could-have and Won't-have stories stay visible in the backlog so that their exclusion is explicit and reversible. This creates a direct traceability chain from the selected ideation concept to the user-story backlog: the report shows which ideas were translated into the MVP, which were deliberately deferred, and why.

Robustness check. A sensitivity check tested whether Concept C's selection depended on one exact set of subjective weights. Transferring five percentage points from research evidence fit to feasibility (both 25%) left Concept C first with 4.25/5, against 4.15/5 for Concepts A and B. Fully reversing the two weights (research evidence fit 20%, feasibility 30%) moved Concept A to 4.20/5, with Concepts B and C at 4.15/5. The selection is therefore robust to a moderate change in priorities, but not to an extreme feasibility-first position. This boundary identifies the precise project risk that the MoSCoW scope and early prototype walkthroughs must control.

5 Product Backlog Structure

5.1 Epic Overview

The backlog is organised into four Epics that follow the shape of the selected Trust-to-Agreement Co-living Journey. Two Epics deliver the MVP journey before and after move-in, while the remaining two hold extension work together with the secondary property management scope so that the first sprint stays focused. Every story carries a stable identifier, which means a later change of sprint or priority does not break its link to the matching use case in Appendix A.

Table 13. Epic structure of the product backlog.

Epic ID	Epic name	Epic goal and value	Linked user stories
EP-01	Housemate Discovery and Trust	Help accommodation seekers judge whether a listing or a potential housemate can be trusted before any commitment is made.	US-09, US-11, US-12, US-13
EP-02	Shared Household Living	Make household expectations explicit and keep everyday responsibilities visible after move-in.	US-01 to US-08 and US-10
EP-03	Accountability Extensions	Strengthen follow-through on accepted responsibilities once the core journey is validated.	US-14, US-15
EP-04	Property Management	Give property managers a direct channel to publish listings and communicate with their tenants.	US-16, US-17, US-18

5.2 User Stories

Every story keeps the standard format that names the user before stating what they want and why. The scope column shows whether a story belongs to the MVP or is deferred to the extension or secondary streams, consistent with the MoSCoW result in Table 12. Tables 14 to 17 list the stories under their Epics.

Table 14. EP-01 Housemate Discovery and Trust user stories.

Story ID	User story	Scope	Jira ticket links
US-09	As an international student new to Australia and unfamiliar with who to trust, I want to communicate with potential housemates securely through the app before sharing my personal contact details, so that I can evaluate compatibility and build trust without exposing myself to scams or harassment during early interactions.	MVP	ST4-35
US-11	As an international student new to Australia with limited local knowledge, I want to view verified housemate profiles with clear identity confirmation, so that I can feel confident I am connecting with a real and trustworthy person before agreeing to share a living space with them.	MVP	ST4-38
US-12	As a social worker with a demanding and irregular work schedule seeking shared accommodation, I want to filter potential housemates by lifestyle preferences such as cleanliness standards, sleep schedule, and daily routine, so that I can find someone whose habits are compatible with mine before committing to living together.	MVP	ST4-39
US-13	As a domestic student moving out of home for the first time, I want to search and filter property listings by location and affordability, so that I can find suitable and trustworthy housing near my university without being overwhelmed or misled by inaccurate listings.	MVP	ST4-43

Table 15. EP-02 Shared Household Living user stories.

Story ID	User story	Scope	Jira ticket links
US-01	As a domestic student managing shared living for the first time, I want to set up a rotating chore roster and track completion history, so that household responsibilities are distributed fairly and I can refer back to who has and hasn't been contributing.	MVP	ST4-23
US-02	As an international student living in a shared apartment for the first time, I want to see clear written rules in the app about things like cleaning the kitchen, taking out rubbish, and keeping noise down, so that I know exactly what I'm expected to do without needing to ask my housemates.	MVP	ST4-24
US-03	As a domestic student living in shared accommodation for the first time, I want to propose a structured chore system through the app, so that I can establish clear and fair expectations with my housemates without needing a confrontational discussion.	MVP	ST4-26
US-04	As a remote worker living in shared accommodation, I want to log shared expenses like groceries, electricity bills, and cleaning supplies with details of who paid, so that I can keep track of household spending and split costs fairly between housemates.	MVP	ST4-28
US-05	As an international student living in shared accommodation, I want to track outstanding balances with my housemates, so that I know exactly who owes me money and how much without having to ask.	MVP	ST4-29
US-06	As an international student sharing a house in Australia, I want to split utility bills such as electricity, water, and internet with my housemates, so that I can ensure I only pay my fair share and avoid confusion with local billing systems.	MVP	ST4-32
US-07	As an avid traveller currently living in shared accommodation, I want a household noticeboard to post updates and suggestions, so that I can communicate with housemates without it getting lost in a busy group chat while I'm on the move.	MVP	ST4-33
US-08	As a domestic student living away from home for the first time, I want a way to raise household issues anonymously through the app, so that I can address uncomfortable problems like bathroom usage or noise without damaging relationships with the people I live with.	MVP	ST4-34
US-10	As a musical artist living in shared accommodation who regularly plays music at high volumes, I want to set and view agreed house rules within the app, so that boundaries around noise and shared spaces are clearly established upfront and I can refer back to them to resolve disputes without confrontation.	MVP	ST4-36

Table 16. EP-03 Accountability Extensions user stories.

Story ID	User story	Scope	Jira ticket links
US-14	As a young professional living in shared accommodation, I want to automatically receive in-app reminders when I have an overdue chore or expense, so that I can stay on top of my responsibilities without housemates needing to confront me directly.	Extension	ST4-25
US-15	As a tradesperson frequently relocating for work and seeking reliable shared accommodation, I want to read honest reviews left by previous housemates, so that I can make a well-informed decision about who I choose to live with and avoid potential conflicts before committing to a shared living arrangement.	Extension	ST4-41

Table 17. EP-04 Property Management user stories.

Story ID	User story	Scope	Jira ticket links
US-16	As a property manager, I want to send automated rent reminders and monitor whether each tenant has paid their rent on time, so that I can reduce late payments without needing to manually chase tenants.	Secondary	ST4-30
US-17	As a property manager overseeing multiple student rental properties in Sydney, I want to publish detailed and accurate rental listings on the platform, so that students searching for accommodation can easily find, evaluate, and enquire about my properties directly without the need for third party listing websites.	Secondary	ST4-40
US-18	As a property manager overseeing multiple student rental properties, I want to send announcements and maintenance updates directly to all tenants in a property through the app, so that I can efficiently communicate important information without the delays and inefficiencies of individual emails or phone calls.	Secondary	ST4-37

6 Acceptance Criteria

Every user story in the backlog has measurable acceptance criteria written in the Given When Then form, so each table in this section can be executed directly as a test skeleton. Each story covers the successful path together with at least one error scenario and one edge scenario. The same criteria are recorded on the matching Jira ticket, whose key appears in each caption so that the backlog in Jira and this report stay verifiable against each other. The tables are grouped under the Epics defined in Table 13.

6.1 EP-01 Housemate Discovery and Trust

Table 18. Acceptance criteria for US-09 (Jira ST4-35).

Acceptance ID	Scenario	Given	When	Then
AC-US09-01	Success	two users are connected on the platform	they open a conversation	they can send and receive messages through a secure in-app system without either party's personal contact details being visible
AC-US09-02	Success	both users mutually agree to share contact details	they both opt in	the system exchanges their contact details between them
AC-US09-03	Success	a user feels unsafe	they block or report another user from the messaging interface	the reported user is flagged for review and the conversation is ended
AC-US09-04	Error	a user attempts to include personal contact details in a message	the system detects this	it warns the user and recommends using the opt-in contact sharing feature instead
AC-US09-05	Error	a user attempts to contact someone who has blocked them	they try to send a message	the system prevents the message from being sent without revealing that they have been blocked
AC-US09-06	Edge case	a user's account is flagged as unverified	they attempt to initiate a conversation	the system displays a warning to the recipient that the sender is unverified before they choose to respond

Table 19. Acceptance criteria for US-11 (Jira ST4-38).

Acceptance ID	Scenario	Given	When	Then
AC-US11-01	Success	a user has submitted a valid form of identification	the verification is approved	their profile displays a verified identity badge visible to all other users
AC-US11-02	Success	a user is searching for housemates	they apply the verified only filter	only profiles with a verified identity badge appear in search results
AC-US11-03	Success	a user views an unverified profile	they open it	a visible warning indicator is displayed clearly at the top of the profile
AC-US11-04	Error	a user submits an expired or unreadable identification document	the system processes it	the verification is rejected and the user is notified with instructions to resubmit a valid document
AC-US11-05	Error	a user attempts to access the platform's core matching features without completing verification	they try to proceed	the system prompts them to complete verification first
AC-US11-06	Edge case	a verified user's identification document expires	the system detects the expiry	it notifies the user to resubmit updated documentation to maintain their verified status

Table 20. Acceptance criteria for US-12 (Jira ST4-39).

Acceptance ID	Scenario	Given	When	Then
AC-US12-01	Success	a user has completed their lifestyle preference profile	they search for housemates	results are automatically ranked by a compatibility score calculated from matching preferences
AC-US12-02	Success	a user wants to evaluate a potential housemate	they open their profile	a detailed compatibility summary is displayed showing areas of strong match and potential differences
AC-US12-03	Success	a potential housemate views the user's profile	they open it	they receive a notification that someone has viewed their profile
AC-US12-04	Error	a user has not completed their lifestyle preference profile	they attempt to search for compatible housemates	the system prompts them to complete their profile before search results can be personalised
AC-US12-05	Error	a user sets contradictory preferences (e.g. early riser and late sleeper simultaneously)	they attempt to save	the system flags the conflict and prompts them to review their selections
AC-US12-06	Edge case	no housemates match a user's strict preference filters	the search returns zero results	the system suggests broadening one or more filters and displays the closest available matches

Table 21. Acceptance criteria for US-13 (Jira ST4-43).

Acceptance ID	Scenario	Given	When	Then
AC-US13-01	Success	a student enters their university name as a reference point	they search for listings	results are displayed ranked by proximity to the selected campus
AC-US13-02	Success	a student applies filters for maximum weekly rent, distance from campus, number of bedrooms and move-in date	they search	only listings matching all applied filters are displayed
AC-US13-03	Success	a student finds a listing of interest	they open it	they can view affordability indicators including weekly rent in AUD, estimated cost per person, a comparison to the local suburb average and a Budget/Moderate/Premium affordability rating
AC-US13-04	Error	a student searches for listings in an area with no available results	the search returns empty	the system notifies them and suggests broadening their search radius or adjusting filters
AC-US13-05	Error	a listing contains inaccurate photos or descriptions	a student reports it	the listing is flagged for review and temporarily hidden from search results if multiple reports are received
AC-US13-06	Edge case	a student's preferred university is not listed in the system	they search	the system allows them to enter a custom address or suburb as an alternative reference point

6.2 EP-02 Shared Household Living

Table 22. Acceptance criteria for US-01 (Jira ST4-23).

Acceptance ID	Scenario	Given	When	Then
AC-US01-01	Success	a user has created a household and added housemates	they set up a chore roster with task assignments and a rotation schedule	the system distributes chores evenly and notifies each housemate of their assigned tasks
AC-US01-02	Success	a housemate has been assigned a chore	they mark it as complete	the system logs the completion with a timestamp and updates the participation summary
AC-US01-03	Success	a user wants to review contributions	they access the chore history	they can see a full log of who completed what and when
AC-US01-04	Error	a user attempts to create a roster without assigning all housemates	they submit	the system prompts them to assign at least one task to each housemate before activating
AC-US01-05	Error	a chore deadline has passed without being marked complete	the system checks for overdue tasks	it flags the chore as overdue and notifies the assigned housemate
AC-US01-06	Edge case	a housemate leaves the household	their assigned chores become unassigned	the system alerts the house admin to reassign those tasks

Table 23. Acceptance criteria for US-02 (Jira ST4-24).

Acceptance ID	Scenario	Given	When	Then
AC-US02-01	Success	a housemate has added household rules and expectations	an international student opens the household rules section	they can view all rules in a single clearly organised list
AC-US02-02	Success	a user has set their language preference to a non-English language	they view household rules	the rules are displayed in their preferred language or simplified English
AC-US02-03	Success	a user wants clarification on a specific rule	they tap on it	they can view an expanded description or explanation provided by the housemate who created it
AC-US02-04	Error	a user attempts to save a rule without a title or description	they submit	the system prompts them to complete all required fields before saving
AC-US02-05	Error	no household rules have been added yet	a user opens the rules section	the system displays a prompt encouraging housemates to add expectations
AC-US02-06	Edge case	a rule is written in English but the user's preferred language is different	the translation service is unavailable	the system displays the rule in simplified English as a fallback

Table 24. Acceptance criteria for US-03 (Jira ST4-26).

Acceptance ID	Scenario	Given	When	Then
AC-US03-01	Success	a user wants to establish a chore system	they submit a proposed chore system including task assignments and schedules	all housemates receive a notification to review the proposal
AC-US03-02	Success	all housemates have reviewed the proposal	they all approve it	the chore system becomes active and assignments are distributed accordingly
AC-US03-03	Success	a housemate wants to suggest changes	they submit feedback on the proposal	the proposer receives a notification and can revise and resubmit
AC-US03-04	Error	a housemate does not respond to the proposal within 7 days	the deadline passes	the system sends a reminder and flags the proposal as awaiting response
AC-US03-05	Error	a user submits a proposal without assigning tasks to all housemates	they attempt to submit	the system prompts them to assign at least one task per housemate
AC-US03-06	Edge case	one housemate rejects the proposal	the rejection is submitted	the system notifies the proposer and pauses activation until a revised proposal is approved by all

Table 25. Acceptance criteria for US-04 (Jira ST4-28).

Acceptance ID	Scenario	Given	When	Then
AC-US04-01	Success	a user has paid for a shared expense	they log it with a description, amount, category and payer	the system automatically calculates each selected housemate's share and notifies them
AC-US04-02	Success	an expense has been logged	a housemate views their dashboard	they can see all expenses they are involved in and their individual amount owed
AC-US04-03	Success	a housemate has paid their share	the payer marks it as settled	the balance between those two housemates updates automatically
AC-US04-04	Error	a user submits an expense without entering an amount	they attempt to save	the system prompts them to complete all required fields
AC-US04-05	Error	a user attempts to split an expense among housemates not in their household	they submit	the system only allows splitting among confirmed household members
AC-US04-06	Edge case	a housemate leaves the household with an outstanding balance	their account is removed	the system retains their balance record and flags it as unresolved for the remaining housemates

Table 26. Acceptance criteria for US-05 (Jira ST4-29).

Acceptance ID	Scenario	Given	When	Then
AC-US05-01	Success	expenses have been logged between housemates	a user opens their balance dashboard	they can see a clear summary of who owes them money and how much
AC-US05-02	Success	a housemate has paid their share	the user marks the balance as settled	the dashboard updates automatically to reflect the new balance
AC-US05-03	Success	balances exist across multiple housemates	the user views the dashboard	all balances are displayed individually per housemate
AC-US05-04	Error	a user attempts to mark a balance as settled that has already been settled	they submit	the system notifies them that the balance is already cleared
AC-US05-05	Error	no expenses have been logged yet	a user opens the balance dashboard	the system displays a prompt to add the first shared expense
AC-US05-06	Edge case	a balance is disputed by a housemate	they flag it	the system marks the balance as disputed and notifies both parties to resolve it before it can be settled

Table 27. Acceptance criteria for US-06 (Jira ST4-32).

Acceptance ID	Scenario	Given	When	Then
AC-US06-01	Success	a utility bill has been received	a user enters the bill type, total amount and billing period	the system calculates and assigns each housemate's share based on the selected split method
AC-US06-02	Success	a bill has been logged	all housemates view the shared bill history	they can see all submitted bills and their individual amounts owed
AC-US06-03	Success	a housemate believes a bill is incorrect	they flag it as disputed	all housemates are notified and the bill is marked as under review
AC-US06-04	Error	a user submits a bill without entering a total amount	they attempt to save	the system prompts them to complete all required fields
AC-US06-05	Error	a user attempts to split a bill among zero housemates	they submit	the system requires at least two participants before allowing submission
AC-US06-06	Edge case	a utility bill covers a period during which a housemate was absent	the bill is split	the system allows the user to manually exclude or adjust that housemate's share for that billing period

Table 28. Acceptance criteria for US-07 (Jira ST4-33).

Acceptance ID	Scenario	Given	When	Then
AC-US07-01	Success	a user wants to post a notice	they create a notice with a title, description and optional image	it is immediately visible to all housemates on the noticeboard
AC-US07-02	Success	a notice has been posted	a housemate wants to respond	they can leave a limited reaction or reply directly on the notice
AC-US07-03	Success	a notice has an expiry date set	the date passes	the notice is automatically archived and removed from the active noticeboard
AC-US07-04	Error	a user attempts to post a notice without a title	they submit	the system prompts them to add a title before posting
AC-US07-05	Error	a user sets an expiry date in the past	they submit	the system rejects the date and prompts them to select a future date
AC-US07-06	Edge case	a notice receives no interaction for 30 days and has no expiry date set	the system checks for inactive notices	it prompts the creator to either renew or archive the notice

Table 29. Acceptance criteria for US-08 (Jira ST4-34).

Acceptance ID	Scenario	Given	When	Then
AC-US08-01	Success	a user wants to raise an issue	they submit it with the anonymous option selected	the issue is posted to the shared household space without revealing the submitter's name or profile
AC-US08-02	Success	an issue has been posted	housemates view it	they can vote on whether they agree it is a concern
AC-US08-03	Success	all parties have agreed the issue is resolved	the submitter or a majority vote marks it as resolved	the issue is closed and archived
AC-US08-04	Error	a user attempts to submit an issue without a description	they submit	the system prompts them to add a description before posting
AC-US08-05	Error	a user attempts to identify an anonymous submitter through the system	they try to access submitter information	the system does not reveal any identifying information under any circumstance
AC-US08-06	Edge case	an anonymous issue receives no votes or responses within 14 days	the system checks for inactive issues	it sends a reminder to all housemates to review and respond

Table 30. Acceptance criteria for US-10 (Jira ST4-36).

Acceptance ID	Scenario	Given	When	Then
AC-US10-01	Success	a user wants to propose a house rule	they submit it with a title, description and category	all housemates receive a notification to review and acknowledge it
AC-US10-02	Success	all housemates have acknowledged a proposed rule	the last acknowledgement is submitted	the rule is officially added to the household rules list with a timestamp
AC-US10-03	Success	a user wants to reference a rule during a dispute	they link it to a household issue	the rule is displayed alongside the issue for all housemates to see
AC-US10-04	Error	a user attempts to submit a rule without a title or category	they submit	the system prompts them to complete all required fields
AC-US10-05	Error	a housemate attempts to edit or remove a rule without majority approval	they submit the change	the system rejects it and notifies all housemates that a change has been requested pending approval
AC-US10-06	Edge case	a rule has not been acknowledged by one housemate after 7 days	the system checks for pending acknowledgements	it sends a reminder to that housemate to review and acknowledge the rule

6.3 EP-03 Accountability Extensions

Table 31. Acceptance criteria for US-14 (Jira ST4-25).

Acceptance ID	Scenario	Given	When	Then
AC-US14-01	Success	a chore deadline has passed and the chore has not been marked complete	the system checks for overdue tasks	it automatically sends an in-app notification to the assigned housemate
AC-US14-02	Success	the house admin wants to manually remind a housemate	they send a nudge	the assigned housemate receives an in-app notification without any personal contact details being exchanged
AC-US14-03	Success	a housemate marks an overdue chore as complete after receiving a reminder	they submit the completion	the system updates the chore log and removes the overdue flag
AC-US14-04	Error	a housemate has disabled notifications	a chore becomes overdue	the system still flags the chore as overdue within the app and displays it prominently on their dashboard
AC-US14-05	Edge case	a housemate is temporarily removed from the roster (e.g. on leave)	their chore deadline passes	the system does not send them an overdue reminder and reassigns the task to the next available housemate

Table 32. Acceptance criteria for US-15 (Jira ST4-41).

Acceptance ID	Scenario	Given	When	Then
AC-US15-01	Success	a confirmed tenancy period has ended	a user submits a written review and star rating for their previous housemate	the review is displayed on that housemate's profile in chronological order
AC-US15-02	Success	a user views a housemate's profile	they scroll to the reviews section	they can see all reviews with the overall star rating displayed prominently at the top
AC-US15-03	Success	a housemate receives a review	it is posted	they receive a notification and can submit a single written response
AC-US15-04	Error	a user attempts to submit a review before a confirmed tenancy period has ended	they try to submit	the system prevents submission and notifies them that reviews can only be left after a confirmed tenancy
AC-US15-05	Error	a user attempts to submit a second review for the same housemate from the same tenancy	they submit	the system rejects the duplicate and notifies the user
AC-US15-06	Edge case	a review is flagged as inappropriate by multiple users	the system detects multiple reports	the review is temporarily hidden pending platform review and the reviewer is notified

6.4 EP-04 Property Management

Table 33. Acceptance criteria for US-16 (Jira ST4-30).

Acceptance ID	Scenario	Given	When	Then
AC-US16-01	Success	a property manager has set a rent amount and due date for a tenant	the due date approaches	the system automatically sends a rent reminder to that tenant
AC-US16-02	Success	a tenant's payment has been recorded	the property manager views the dashboard	the tenant's status updates to "paid"
AC-US16-03	Success	the due date has passed without payment being recorded	the system checks payment status	the tenant's status updates to "overdue" and the property manager is notified
AC-US16-04	Error	a property manager attempts to set a rent due date without entering a rent amount	they submit	the system prompts them to complete all required fields
AC-US16-05	Error	a tenant's contact details are incomplete	the system attempts to send a reminder	the system flags the tenant's profile as incomplete and alerts the property manager
AC-US16-06	Edge case	a tenant disputes a rent amount	they flag it through the app	the system marks the payment status as disputed and notifies the property manager to review

Table 34. Acceptance criteria for US-17 (Jira ST4-40).

Acceptance ID	Scenario	Given	When	Then
AC-US17-01	Success	a property manager has completed a listing with photos, description, rent price, available date and house rules	they publish it	it immediately appears in student search results based on location and filters
AC-US17-02	Success	a student finds a listing they are interested in	they send an enquiry	the property manager receives a notification and the message is stored in their centralised inbox
AC-US17-03	Success	a listing's available date has passed	the system checks active listings	the listing is automatically removed from search results and the property manager is notified
AC-US17-04	Error	a property manager attempts to publish a listing without photos or a rent price	they submit	the system prompts them to complete all required fields before publishing
AC-US17-05	Error	a student attempts to enquire about a listing that has already been filled	they submit the enquiry	the system notifies them that the listing is no longer available
AC-US17-06	Edge case	a listing has been live for 60 days without being filled	the system checks listing age	it prompts the property manager to review and update the listing details

Table 35. Acceptance criteria for US-18 (Jira ST4-37).

Acceptance ID	Scenario	Given	When	Then
AC-US18-01	Success	a property manager wants to send an announcement	they compose it with a title, description, category and optional attachment	it is delivered to all tenants in the selected property
AC-US18-02	Success	an announcement has been sent	a tenant opens the app	they receive a push notification and can view the announcement in their notification centre
AC-US18-03	Success	a property manager wants to schedule a recurring announcement	they set a recurrence schedule	the system automatically sends the announcement at the defined intervals
AC-US18-04	Error	a property manager attempts to send an announcement without selecting a property	they submit	the system prompts them to select at least one property before sending
AC-US18-05	Error	a scheduled announcement fails to send	the system detects the failure	it retries delivery and notifies the property manager if the issue persists
AC-US18-06	Edge case	a tenant has not viewed a high priority announcement within 24 hours	the system checks read receipts	it sends a follow-up push notification to that tenant

7 Functional and Non-Functional Requirements

7.1 Functional Requirements

Required content

Define what the system must do. Link every requirement to one or more user stories and state how the requirement will be verified.

Table 36. Functional requirements template.

Requirement ID	Functional requirement	Related stories	Verification method
FR-01	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-02	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-03	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>

7.2 Non-Functional Requirements

Required content

Include measurable requirements for privacy, security, accessibility, performance and usability. Add other categories where they are relevant to the solution.

Table 37. Non-functional requirements template.

Requirement ID	Category	Measurable requirement	Verification method
NFR-01	Privacy	<i>[Privacy target]</i>	<i>[Verification method]</i>
NFR-02	Security	<i>[Security target]</i>	<i>[Verification method]</i>
NFR-03	Accessibility	<i>[Accessibility target]</i>	<i>[Verification method]</i>
NFR-04	Performance	<i>[Performance target]</i>	<i>[Verification method]</i>
NFR-05	Usability	<i>[Usability target]</i>	<i>[Verification method]</i>

8 User Story Dependencies

Each user story maps one to one onto a goal-level use case, so the dependency analysis is presented with use case identifiers and applies directly to the corresponding stories. The analysis also covers four supporting use cases together with two seeded data sources, because they enable several stories at once rather than belonging to a single story. The full mapping between stories and use cases is recorded in Appendix A.

8.1 Dependency Types

Not every dependency blocks development in the same way, so each relationship is classified before it is used for sprint planning. Table 38 defines the types used in the analysis.

Table 38. Dependency types used in the analysis.

Type	Meaning
Required	The dependent use case cannot proceed until the prerequisite is satisfied.
Required data	The dependent use case needs data that the prerequisite creates or maintains.
Required data (OR)	Any one of the listed prerequisites supplies the needed data.
Conditional	The dependency applies only in a specific situation, such as showing a verified badge.
Optional	The relationship adds support but is not required for the main success path.

8.2 Dependency Table

Tables 39 to 41 group the dependencies by Epic so that the build order within each part of the journey stays visible. The supporting prerequisite SUP-02 appears repeatedly because household-scoped features cannot operate without a confirmed membership.

Table 39. Dependencies within EP-02 Shared Household Living.

Dependent use case	Depends on	Type	Reason
UC-01 Chore roster	SUP-02 Household membership	Required	Only members of an active household can view or manage its chore roster.
UC-02 View household rules	SUP-02 Household membership	Required	Rules belong to a specific household and only members can view or manage them.
UC-02 View household rules	UC-10 Agreed house rules	Required data	Rules must be set and agreed before they can be displayed in a member's preferred language.
UC-03 Propose chore system	UC-01 Chore roster	Required data	A structured chore system cannot be proposed or activated without an existing roster to build on.
UC-03 Propose chore system	SUP-02 Household membership	Required	All approving housemates must be identified members of the same household.
UC-04 Log and split expenses	SUP-02 Household membership	Required	Expenses and each member's share must belong to an authorised household.
UC-05 Track balances	UC-04 Shared expenses	Required data	Balances are computed from logged expenses, so at least one expense must exist to show who owes what.
UC-06 Split utility bills	UC-04 Shared expenses	Required	A utility bill is itself a shared expense, so UC-06 reuses the logging and cost-splitting capability of UC-04 rather than any pre-existing expense data.
UC-07 Household notices	SUP-02 Household membership	Required	The noticeboard is scoped to a single household and only members can post or read notices.
UC-08 Anonymous issues	SUP-02 Household membership	Required	The issue must belong to the reporter's household while hiding their identity from other residents.
UC-08 Anonymous issues	UC-10 Agreed house rules	Optional	Existing house rules give issues a shared reference point during disputes, but an issue can be raised without them.

Table 40. Dependencies within EP-01 Housemate Discovery and Trust.

Dependent use case	Depends on	Type	Reason
UC-09 Secure messaging	UC-11 Verified profiles	Optional	Core secure messaging works on its own with personal contact details hidden, while UC-11 adds the flag that marks unverified counterparties for extra caution.
UC-11 Verified profiles	SUP-04 Verification status or DATA-02 Seeded profiles	Required data (OR)	A stored verification status or a seeded sample profile must exist before a badge can be shown.
UC-12 Lifestyle filtering	UC-11 Verified profiles	Required	Compatibility matching is only offered between verified profiles to preserve trust, and a match does not grant household access.
UC-13 Search listings	UC-17 Managed listings or DATA-01 Seeded listings	Required data (OR)	At least one published listing, whether managed or seeded, must exist before a student can search it.

Table 41. Dependencies within EP-03 Accountability Extensions and EP-04 Property Management.

Dependent use case	Depends on	Type	Reason
UC-14 Overdue reminders	UC-01 Chore roster or UC-04 Shared expenses	Required data (OR)	A reminder needs an overdue chore assignment or an overdue expense to reference.
UC-14 Overdue reminders	SUP-03 Push notifications	Required	Automatic overdue reminders are delivered through the notification service.
UC-15 Housemate reviews	SUP-02 Household or tenancy membership	Required	A review may only be submitted after a confirmed tenancy or shared living arrangement has ended.
UC-16 Rent tracking	SUP-02 Tenancy membership	Required	Rent status needs a confirmed relationship between tenant and property, and selecting an applicant is not enough.
UC-16 Rent tracking	SUP-03 Push notifications	Optional	Push notifications deliver due date and overdue rent reminders outside the active app.
UC-17 Manage listings	SUP-04 Verification status	Conditional	Verification is required only before a verified badge is shown, and an unverified or pending listing may still be saved.
UC-18 Tenant announcements	SUP-02 Tenancy membership	Required	The system needs a confirmed current tenant audience for the selected property.
UC-18 Tenant announcements	SUP-03 Push notifications	Optional	Push notifications deliver the announcement outside the active app.

8.3 Dependency Graph

Figure 3 shows the validated dependency graph for the complete backlog. Solid arrows mark required relationships while dotted arrows mark optional or conditional support. The diamond nodes are OR points where any single source is sufficient, which allows the Sprint 1 prototype to rely on seeded data instead of the secondary property management workflow.

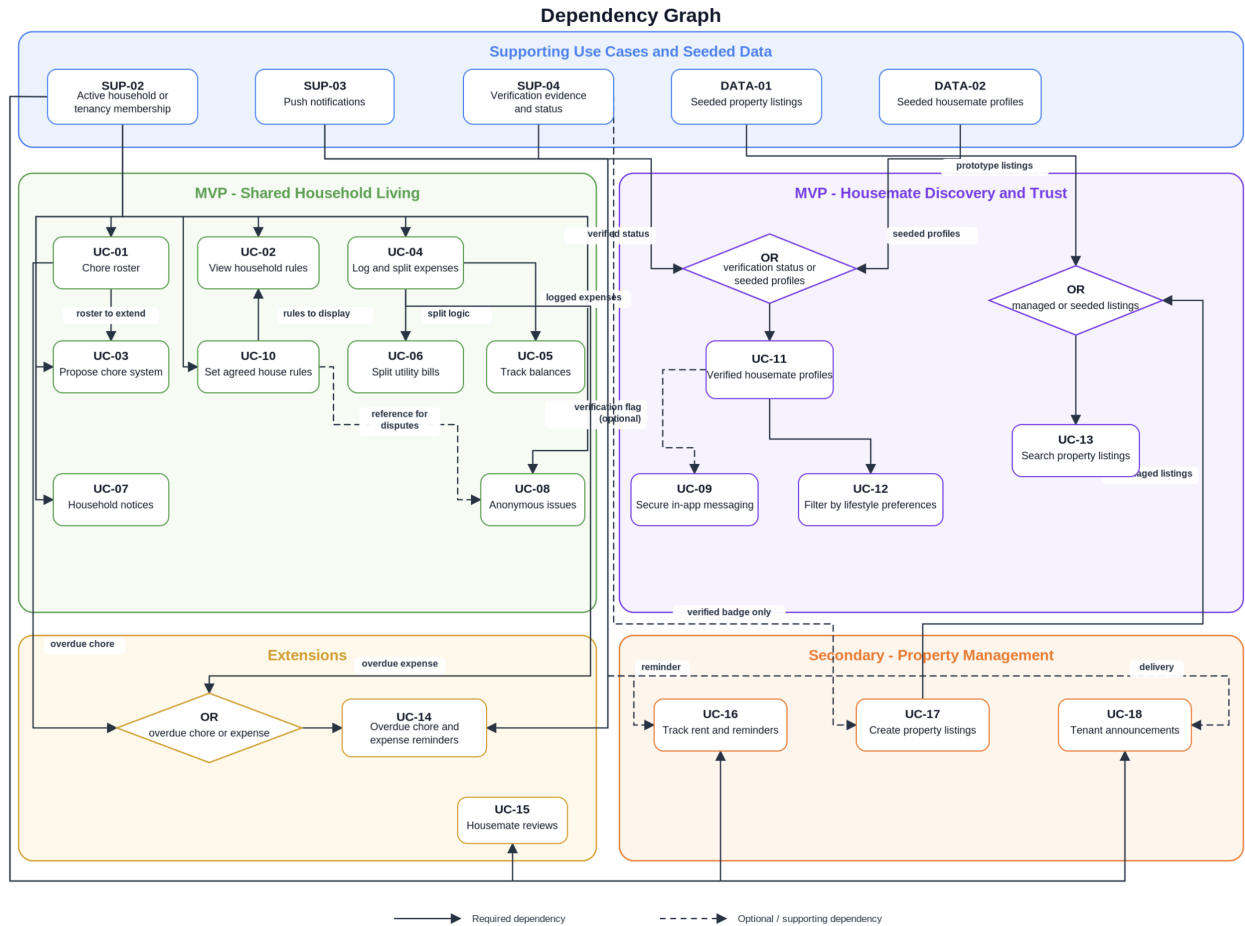


Figure 3. Use case dependency graph for the product backlog.

9 First-Pass User Story Map

Required content

Show the user activities, user tasks and release slices. Clearly identify which stories belong to Sprint 1 and which stories are scheduled for later sprints or the backlog.

Table 42. First-pass user story map template.

Release slice	[Activity 1]	[Activity 2]	[Activity 3]	[Activity 4]	[Activity 5]
User task	[Task]	[Task]	[Task]	[Task]	[Task]
Sprint 1	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]
Later sprint	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]
Backlog	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]

9.1 Sprint 1 Scope and Rationale

Required content

List the Sprint 1 stories and explain why they form a coherent, testable path through the selected concept.

[Insert the approved Sprint 1 story IDs and rationale.]

10 Preliminary Prototype

10.1 Prototype Scope

Required content

The preliminary prototype must correspond to the selected user stories. It must demonstrate the main Sprint 1 path and allow the related features to be tested.

[Describe the prototype scope, target users and main task flow.]

10.2 Prototype Flow

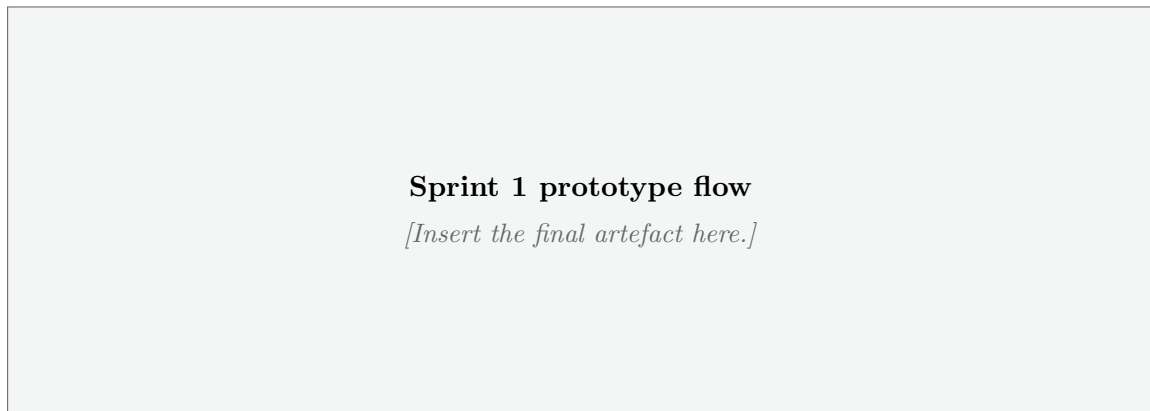


Figure 4. Sprint 1 prototype flow.

10.3 Prototype Screens

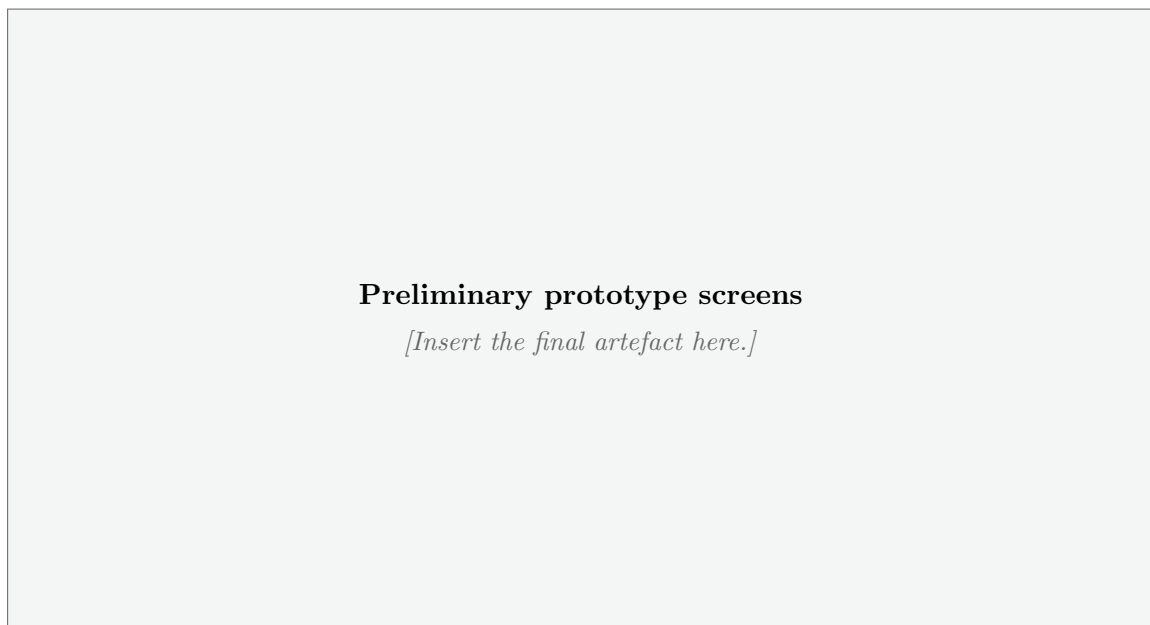


Figure 5. Preliminary prototype screens.

11 Prototype-to-User-Story Traceability

Each screen of the HTML prototype is mapped to the user story and the acceptance criteria that it demonstrates, so every Sprint 1 requirement can be traced from the backlog to an interaction that can be exercised in the prototype. Table 43 records this mapping, and the same screen identifiers are used by the test plan in Section 12. Together the two sections close the chain that starts at a research finding and ends at a testable screen. The link column is completed once the prototype pages are hosted.

Table 43. Prototype-to-user-story traceability.

Screen ID	Prototype screen	Story ID	Acceptance ID	Testable interaction	Link
S-01	Lifestyle profile setup	US-12	AC-US12-04, AC-US12-05	Complete the required preference fields, submit an incomplete form to trigger the completion prompt, save contradictory selections to trigger the conflict flag.	
S-02	Housemate and listing search results	US-13, US-11	AC-US13-01, AC-US13-02, AC-US11-02	Search from a university reference point, apply rent and distance filters, switch on the verified-only filter.	
S-03	Listing and profile detail	US-11, US-13	AC-US11-03, AC-US13-03	Open a verified and an unverified profile to compare the badge with the warning indicator, locate the affordability details.	
S-04	Compatibility comparison	US-12	AC-US12-01, AC-US12-02, AC-US12-06	Compare two candidate profiles through the compatibility summary, apply a strict filter to reach the zero-result broadening suggestion.	
S-05	Guided in-app chat	US-09	AC-US09-01, AC-US09-04, AC-US09-06	Send a first message from a guided prompt, type contact details to trigger the warning, view the unverified sender notice.	
S-06	House rule proposal	US-10	AC-US10-01, AC-US10-04	Propose a rule with a title, description and category, submit with missing fields to trigger the completion prompt.	
S-07	Household rules list and acknowledgement	US-10, US-02	AC-US10-02, AC-US02-01	Acknowledge a proposed rule as each housemate, then find the active rule in the rules list.	

12 Prototype Test Plan

The prototype is validated through five moderated usability tests that follow the Sprint 1 path of the selected concept. Every test is anchored to Sprint 1 user stories and to specific acceptance criteria from Section 6, so a passed test provides direct evidence that the corresponding criteria hold in the prototype. Each test uses five participants with a think-aloud protocol, and every task scenario starts from the prototype screen recorded in Table 43.

Table 44. Test overview and coverage of Sprint 1 stories.

Test	Goal	Stories	Acceptance criteria	Responsible
T1	Lifestyle preference profile setup	US-12	AC-US12-04, AC-US12-05	Pranav
T2	Verification trust signals on listings and profiles	US-11, US-13	AC-US11-02, AC-US11-03, AC-US13-03	Nicky
T3	Compatibility comparison and match choice	US-12	AC-US12-01, AC-US12-02, AC-US12-06	Liam
T4	Guided safe first contact	US-09	AC-US09-01, AC-US09-04, AC-US09-06	Dylan
T5	House rule proposal and acknowledgement	US-10, US-02	AC-US10-01, AC-US10-02, AC-US10-04, AC-US02-01	Liam

Table 45. Test T1. Lifestyle preference profile setup.

Aspect	Description
Test goal	Verify users can complete the lifestyle preference profile without guidance.
Related stories	US-12
Acceptance criteria	AC-US12-04, AC-US12-05
Target user	Student new to shared housing.
Task scenario	“You just signed up. Set up your lifestyle preferences (cleanliness, sleep schedule, noise, guests, smoking).”
Success criteria	$\geq 4/5$ users complete the profile in ≤ 3 minutes with no facilitator help and all required fields filled. An incomplete submission triggers the completion prompt (AC-US12-04). Contradictory selections trigger the conflict flag (AC-US12-05).
Expected errors	Users skip required fields thinking they are optional. Users cannot find where profile setup starts. Field labels are unclear.
Data to collect	Task completion rate, time on task, fields left blank and points where users ask for help.
Responsible	Pranav

Table 46. Test T2. Verification trust signals on listings and profiles.

Aspect	Description
Test goal	Verify users can find a listing’s key details and correctly read the verification status.
Related stories	US-11, US-13
Acceptance criteria	AC-US11-02, AC-US11-03, AC-US13-03
Target user	Student searching for a room.
Task scenario	“Find a room under \$350 per week, check whether the listing and the profile behind it are verified, and open the property details.”
Success criteria	$\geq 4/5$ users correctly state the verification status and find the key details in ≤ 2 minutes. The verified-only filter or badge is identified (AC-US11-02). The unverified warning is noticed and explained (AC-US11-03). The affordability details are located (AC-US13-03).
Expected errors	Users miss the verification badge. Users confuse listing verification with profile verification. Users cannot explain what verified means.
Data to collect	Verification status identification accuracy, time to locate details and the user’s stated meaning of the badge.
Responsible	Nicky

Table 47. Test T3. Compatibility comparison and match choice.

Aspect	Description
Test goal	Verify users can compare housemate compatibility and identify the better match.
Related stories	US-12
Acceptance criteria	AC-US12-01, AC-US12-02, AC-US12-06
Target user	Student choosing between candidate housemates.
Task scenario	“Compare two housemate profiles against your own preferences and pick the better match. Explain your choice.”
Success criteria	$\geq 4/5$ users pick the higher-compatibility profile and cite at least one specific mismatch from the compatibility summary (AC-US12-02). Users correctly interpret the ranked results (AC-US12-01). When a zero-result filter is applied, users notice the broadening suggestion (AC-US12-06).
Expected errors	Users misread the comparison layout. Users judge by photo instead of preference data. The comparison entry point is not found.
Data to collect	Match selection accuracy, reasons given for the choice and misclick count on the comparison screen.
Responsible	Liam

Table 48. Test T4. Guided safe first contact.

Aspect	Description
Test goal	Verify users can send a guided first message without sharing personal contact details.
Related stories	US-09
Acceptance criteria	AC-US09-01, AC-US09-04, AC-US09-06
Target user	International student making first contact.
Task scenario	“You want to ask about this listing. Send a first message using the app.”
Success criteria	$\geq 4/5$ users send a message using at least one guided prompt in ≤ 2 minutes with no contact details visible to either party (AC-US09-01). No user types personal contact details without seeing the warning (AC-US09-04). Users understand the unverified sender warning (AC-US09-06).
Expected errors	Users ignore the prompts and write free text. Users look for a phone number instead of the in-app chat. Prompt wording is unclear to non-native speakers.
Data to collect	Prompt usage rate, task completion rate, attempts to share contact details and comprehension issues noted in think-aloud.
Responsible	Dylan

Table 49. Test T5. House rule proposal and acknowledgement.

Aspect	Description
Test goal	Verify users can propose a house rule and reach an acknowledged agreement.
Related stories	US-10, US-02
Acceptance criteria	AC-US10-01, AC-US10-02, AC-US10-04, AC-US02-01
Target user	Student moving in with new housemates.
Task scenario	“Propose a house rule about quiet hours, have your housemates acknowledge it, then find it in the household rules list.”
Success criteria	$\geq 4/5$ users propose a rule with a title, description and category so that housemates are notified (AC-US10-01), and reach the acknowledged state in ≤ 3 minutes (AC-US10-02). A submission with missing fields triggers the completion prompt (AC-US10-04). The active rule is then found in the rules list (AC-US02-01).
Expected errors	Users cannot find the proposal entry point. Users confuse a proposed rule with an active rule. Users miss the acknowledgement step.
Data to collect	Task completion rate, time on task, drop-offs at the acknowledgement step and comprehension notes from think-aloud.
Responsible	Liam

After each round the team compiles the results into one summary, ranks the observed issues by severity and feeds them into the next prototype iteration. Failed acceptance criteria are recorded by their identifiers so that the retest after iteration targets exactly those criteria.

13 Conclusion

Section purpose

Summarise the selected concept, the Sprint 1 scope, the preliminary prototype and the next validation step. Do not introduce new evidence in the conclusion.

[Insert the conclusion after the report content is complete.]

14 References

- *[Insert references reused from Part 1 and any new sources used in Part 2.]*

A Full User Story and Use Case Mapping

Each user story maps to exactly one goal-level use case, and both identifiers stay stable even when a sprint or priority changes. Table 50 records the complete mapping together with the primary actor and the successful outcome of each use case.

Table 50. User story to use case mapping.

Story	Epic	Use case	Primary actor	Successful outcome	Scope	Website screenshot
US-01	EP-02	UC-01 Set up and manage chore roster	Domestic student / shared-house resident	The user can create, assign and rotate chores among housemates and track completion history.	MVP	
US-02	EP-02	UC-02 View and manage household rules	International student / shared-house resident	The user can view all household rules and expectations in their preferred language in a single location.	MVP	
US-03	EP-02	UC-03 Propose and approve chore system	Domestic student	All housemates have reviewed and approved a structured chore system before it becomes active.	MVP	
US-04	EP-02	UC-04 Log and split shared expenses	Shared-house resident	The user can log a shared expense and the system automatically calculates and assigns each housemate's share.	MVP	
US-05	EP-02	UC-05 Track outstanding balances	International student / shared-house resident	The user can view a real-time summary of who owes what and mark balances as settled.	MVP	
US-06	EP-02	UC-06 Split utility bills	International student / shared-house resident	The user can input a utility bill and split it equally or by usage among selected housemates.	MVP	
US-07	EP-02	UC-07 Post and view household notices	Shared-house resident	The user can post a notice visible to all housemates on a dedicated noticeboard separate from general messaging.	MVP	
US-08	EP-02	UC-08 Raise anonymous household issues	Domestic student / shared-house resident	A household issue is posted to the shared space without revealing the submitter's identity if the anonymous option is selected.	MVP	
US-09	EP-01	UC-09 Send secure in-app messages	International student / shared-house resident	The user can send and receive messages without either party's personal contact details being visible.	MVP	
US-10	EP-02	UC-10 Set and view agreed house rules	Shared-house resident	All housemates have acknowledged a proposed house rule before it becomes active and can reference it during disputes.	MVP	
US-11	EP-01	UC-11 View verified housemate profiles	International student / accommodation seeker	The user can view verified profiles with identity badges and filter search results to show only verified profiles.	MVP	
US-12	EP-01	UC-12 Filter housemates by lifestyle preferences	Shared-house resident / accommodation seeker	The user can set lifestyle preference filters and view a compatibility score and summary for each potential housemate.	MVP	
US-13	EP-01	UC-13 Search and filter property listings	Domestic student / accommodation seeker	The user can search for listings by university proximity and apply filters for rent, distance, bedrooms and move-in date.	MVP	
US-14	EP-03	UC-14 Receive overdue chore and expense reminders	Shared-house resident	The user receives automatic in-app reminders when a chore or expense payment is overdue.	Extension	
US-15	EP-03	UC-15 Read and submit housemate reviews	Shared-house resident / accommodation seeker	The user can read reviews left by previous housemates and submit a review after a confirmed tenancy ends.	Extension	
US-16	EP-04	UC-16 Track rent and send reminders	Property manager	Rent status is recorded per tenant and overdue tenants receive automatic reminders.	Secondary	
US-17	EP-04	UC-17 Create and manage property listings	Property manager	A complete property listing is published and visible to students searching for accommodation.	Secondary	
US-18	EP-04	UC-18 Send tenant announcements	Property manager	An announcement is delivered to all tenants of a selected property with read receipt tracking.	Secondary	

Supporting use cases enable several user goals at once and therefore do not map to a single story. Two seeded data sources let the prototype demonstrate the discovery features without implementing the secondary property workflow first.

Table 51. Supporting use cases.

ID	Supporting use case	Primary actor	Successful outcome
SUP-01	Manage account and role access	Any registered user	The user is authenticated and receives the correct student, resident, or property-manager permissions.
SUP-02	Establish active household or tenancy membership	Resident / property manager / selected tenant	Membership is activated only after a household invitation or tenancy is confirmed.
SUP-03	Deliver push notifications	System	Chore reminders, expense reminders, announcement deliveries, and issue notifications can reach users outside the active app.
SUP-04	Manage verification evidence and status	User / property manager / system	Evidence is stored and the related profile receives a pending, verified, or rejected status.

Table 52. Supporting prototype data.

Data ID	Data source	Purpose
DATA-01	Seeded property listings	Provides published sample listings when the MVP prototype does not implement the secondary property-listing workflow.
DATA-02	Seeded housemate profiles	Provides sample verified and unverified profiles for compatibility matching and verification testing in the MVP prototype.

B Prototype Screens and Links

Appendix content

Insert full-size prototype screenshots, hosted page links and any interaction notes needed for assessment.

C Jira Evidence

Appendix content

Insert Jira backlog exports, screenshots or links showing Epics, user stories, acceptance criteria and sprint assignment.